Numbers, Percents & Ratios

ANSWER KEY

Section 1: Number Properties, Absolute Value & Order of Operations

1 What is the value of the numerical expression $|-14|-3|5-9|+(-2)^3$?

a. -6 ✓

b. -10

c. 2

d. 14

$|-14|=14$, $|5-9|=|-4|=4$, and $(-2)^3=-8$. Substituting gives $14-3(4)+(-8)=14-12-8=-6$.

2 If p is a positive prime number and q is a negative integer, which of the following values must always be negative?

a. $p-q$

b. $q-p$ ✓

c. p^2+q^2

d. $\frac{p}{q^2}$

Since $p>0$ and $q<0$, $-p<0$. Therefore, $q-p$ is the sum of two negative numbers, which is always negative.

3 What is the greatest common factor of $72, 108,$ and 180?

a. 18

b. 24

c. 36 ✓

d. 72

Prime factorizations: $72=2^3\times3^2$, $108=2^2\times3^3$, $180=2^2\times3^2\times5$. GCF $=2^2\times3^2=4\times9=36$.

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Section 2: Proportions, Rates & Scale Conversions

4 On a blueprint, a scale of $\frac{1}{4}$ inch represents 3 feet. If a conference room has a length of 36 feet, what is its length on the blueprint?

a. 2.5 inches

b. 3 **inches** ✓

c. 4.5 inches

d. 6 inches

Set up the proportion: $\frac{0.25 \text{ in}}{3 \text{ ft}} = \frac{x \text{ in}}{36 \text{ ft}}$. Cross-multiplying yields $3x = 0.25 \times 36 = 9 \Rightarrow x = 3$ inches.

5 A car travels 180 miles on 6 gallons of fuel. At the same rate of fuel consumption, how many gallons of fuel are required to travel 450 miles?

a. 12

b. 15 ✓

c. 16.5

d. 18

Fuel efficiency is $\frac{180}{6} = 30$ miles per gallon. For 450 miles: $\frac{450}{30} = 15$ gallons.

6 The ratio of copper to zinc in a brass alloy is $5 : 3$ by weight. If a piece of brass weighs 56 kilograms, how many kilograms of copper does it contain?

a. 21

b. 30

c. 35 ✓

d. 40

Total ratio parts $= 5 + 3 = 8$. Each part weighs $\frac{56}{8} = 7$ kg. Weight of copper $= 5 \times 7 = 35$ kg.

Section 3: Percentages, Discounts & Successive Changes

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7 A store purchases a bicycle for 150 and marks up the price by 40%. During a weekend clearance sale, the bicycle is discounted by 20%. What is the final sale price?

a. 168 ✓

b. 174

c. 180

d. 192

Retail price after 40% markup $= 150 \times 1.40 = 210$. After 20% discount, final price $= 210 \times 0.80 = 168$.

8 If 35% of a number N is equal to 84, what is 60% of N?

a. 120

b. 144 ✓

c. 168

d. 240

$0.35N = 84 \Rightarrow N = \frac{84}{0.35} = 240$. Then 60% of $240 = 0.60 \times 240 = 144$.

9 The price of an airline ticket increased from 250 to 320. What was the percent increase in the ticket price?

a. 22%

b. 25%

c. 28% ✓

d. 30%

Increase $= 320 - 250 = 70$. Percent increase $= \frac{70}{250} \times 100\% = \frac{7}{25} \times 100\% = 28\%$.

Section 4: Scientific Notation, Exponents & Radicals

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10 What is the value of $(3.2 \times 10^{7}) \times (2.5 \times 10^{-3})$ expressed in standard scientific notation?

a. 8.0×10^{4} ✓

b. 8.0×10^{5}

c. 8.0×10^{3}

d. 0.8×10^{5}

Multiply coefficients: $3.2 \times 2.5 = 8.0$. Add exponents: $10^{7+(-3)} = 10^{4}$. The result is 8.0×10^{4}.

11 Which of the following is equivalent to $\frac{x^{5} \cdot x^{-2}}{(x^{3})^{2}}$ for all positive values of x?

a. x^{-3} ✓

b. x^{-1}

c. x^{0}

d. x^{2}

Numerator $= x^{5-2} = x^{3}$. Denominator $= x^{3 \times 2} = x^{6}$. Quotient $= \frac{x^{3}}{x^{6}} = x^{3-6} = x^{-3}$.

12 Simplify the radical expression $\sqrt{75} - 2\sqrt{27} + \sqrt{48}$.

a. $2\sqrt{3}$

b. $3\sqrt{3}$ ✓

c. $4\sqrt{3}$

d. $5\sqrt{3}$

Simplify each radical: $\sqrt{75} = 5\sqrt{3}$, $2\sqrt{27} = 2(3\sqrt{3}) = 6\sqrt{3}$, and $\sqrt{48} = 4\sqrt{3}$. Combining yields $(5 - 6 + 4)\sqrt{3} = 3\sqrt{3}$.

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